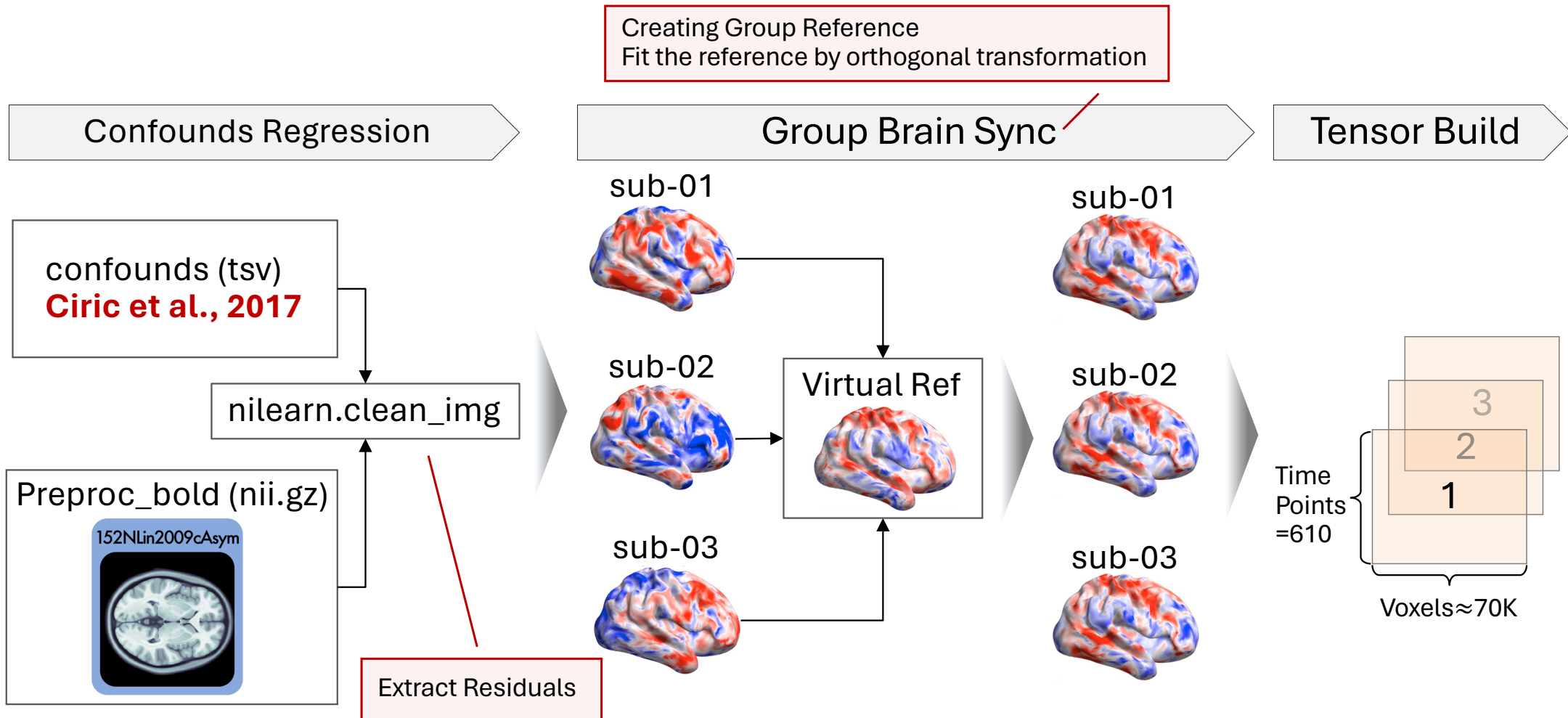
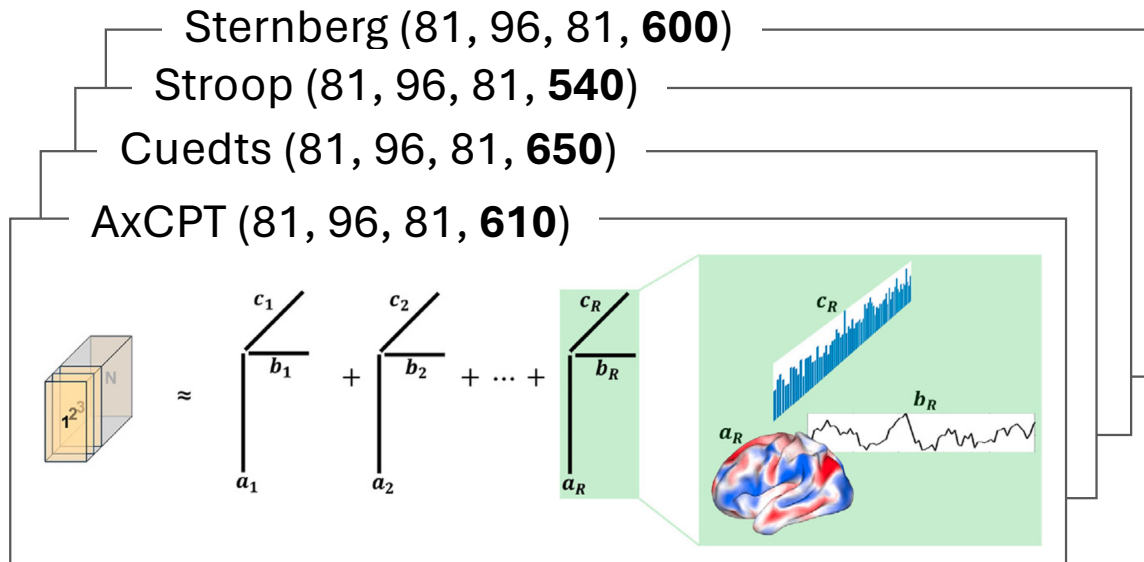


Analysis 1/3

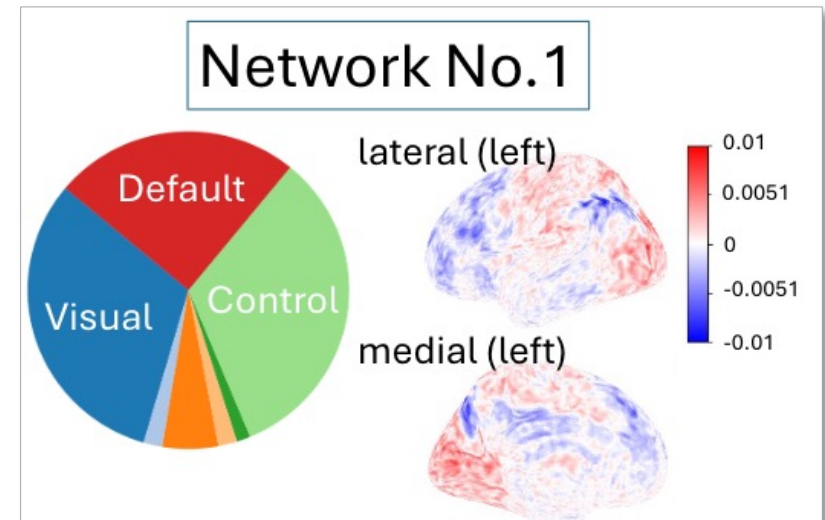


Analysis 2/3

① Tensor Decomposition for Each Task



② Network Ratio Extraction



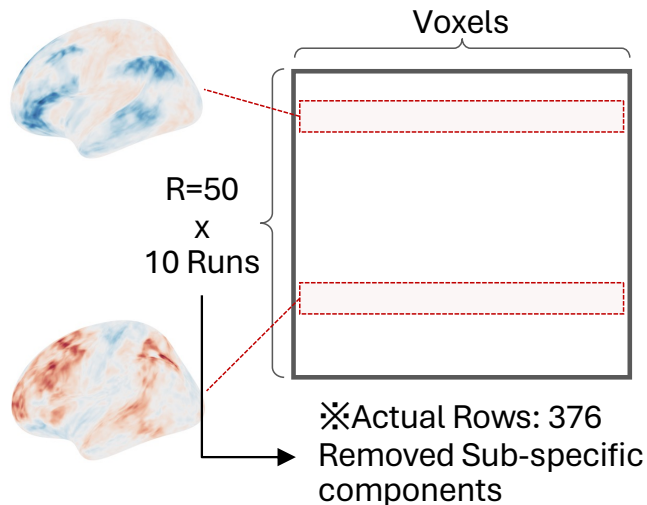
Analysis 3/3

Aim

Identify the brain network represented by the topological mode of each component

Tensor Decomposition

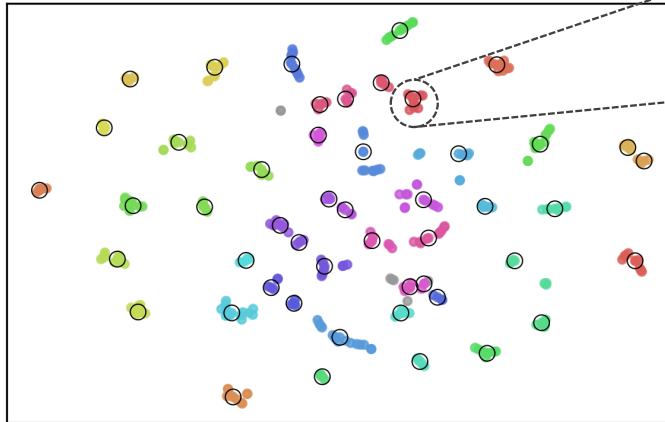
- Tensor Decomposition for Each Task (10 Runs w/ Random Initialization)
- 50 Components x 10 runs into matrix



Manifold Learning (t-SNE)

- Manifold learning and a 2D spatial map
- HDBSCAN for clustering

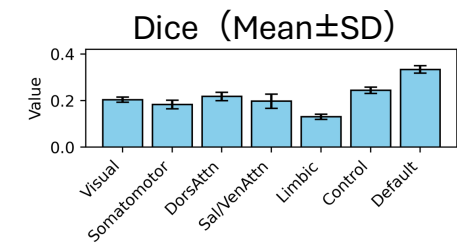
Perplexity = 3 Clusters: 49 Noise: 7



Network Composition by Cluster

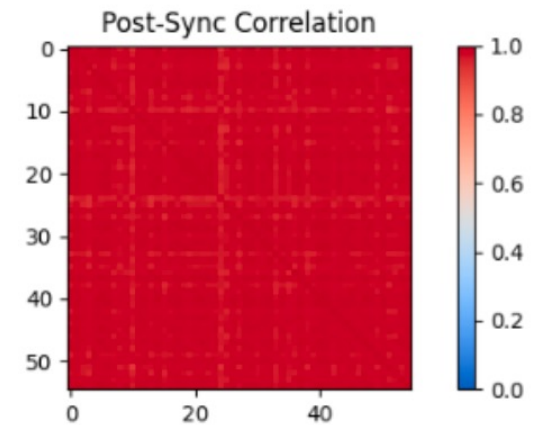
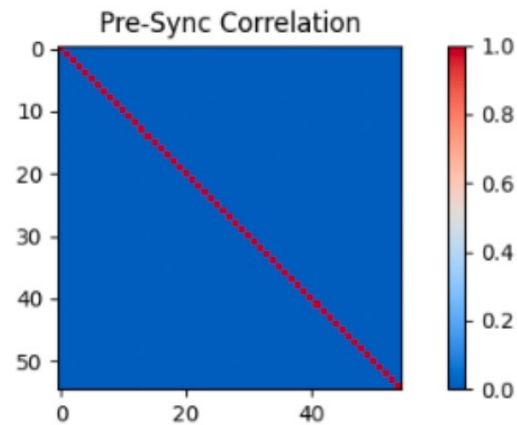
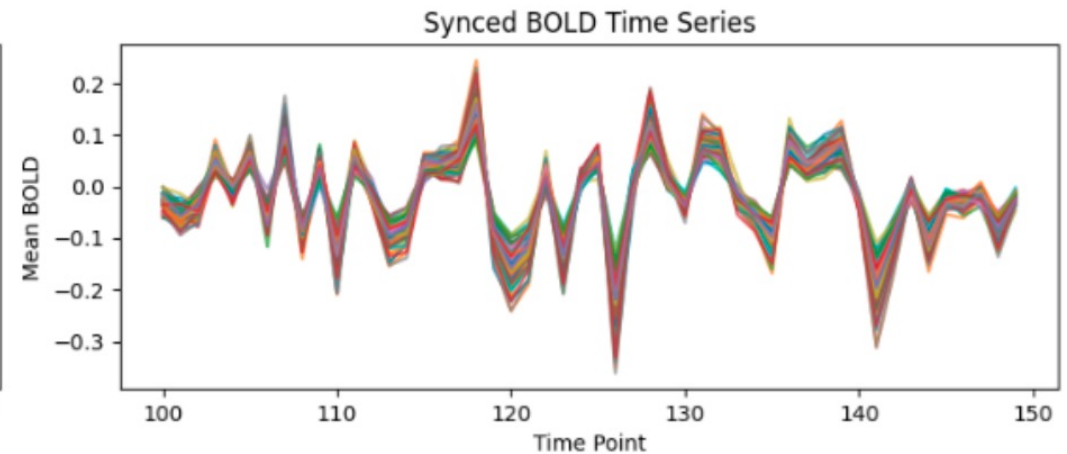
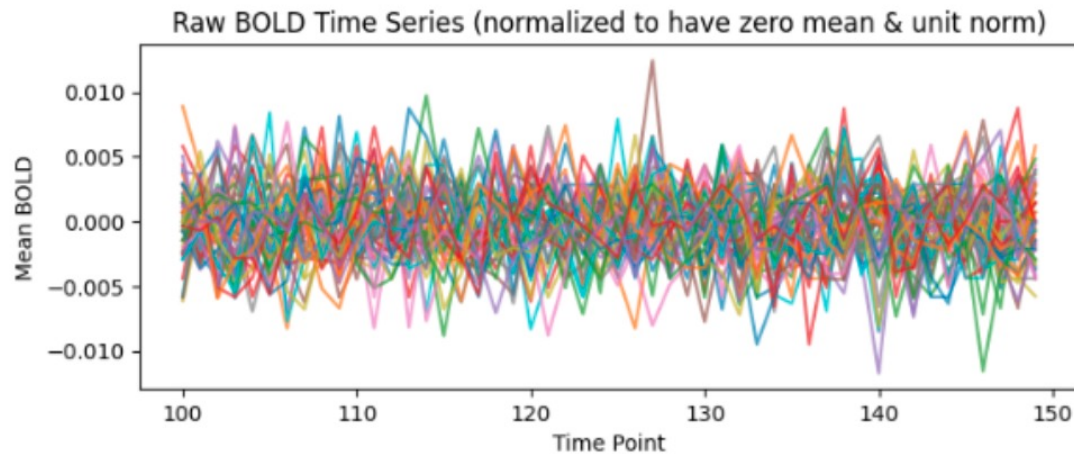
- Calculate the proportion of each network within each cluster
- Identify networks with statistically significant representation.

Net	Cluster 24						
	Visual	Somatomotor	DorsAttn	Sal/VenAttn	Limbic	Control	Default
axcpt_r1_48	0.2168	0.1713	0.2128	0.2224	0.1216	0.2460	0.3468
axcpt_r5_48	0.1953	0.1684	0.1891	0.2065	0.1156	0.2691	0.3563
axcpt_r6_50	0.2019	0.1770	0.2336	0.2050	0.1460	0.2434	0.3322
axcpt_r7_49	0.2184	0.1830	0.2161	0.2162	0.1312	0.2269	0.3250
axcpt_r10_50	0.1896	0.2192	0.2394	0.1384	0.1398	0.2389	0.3111



Results of Group Brain Sync

Normalize the signals, average them across time points, and compute the correlation matrix. An increase in correlation coefficients confirms the effectiveness of Brain Sync.



Results of Spatial Mapping (e.g., AX-CPT)

